Summary in Graph

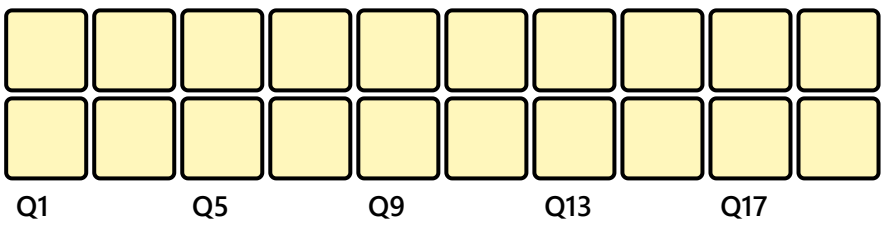
Exam Summary (GO Classes GATE CS | Calculus | Weekly Quiz 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	20 (10 + 10)
Total Marks:	30 (10 + 20)
Exam Duration:	60 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (14) out of 138 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Calculus

Suppose that f is continuous on $[a, b]$ and differentiable on (a, b). If $f'(x) > 0$ on (a, b). Which of the following is necessarily true?

A. f is decreasing on $[a, b]$,
B. f has no local extrema on (a, b),
C. f is a constant function on (a, b),

D. f is concave up on (a, b),

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 02sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

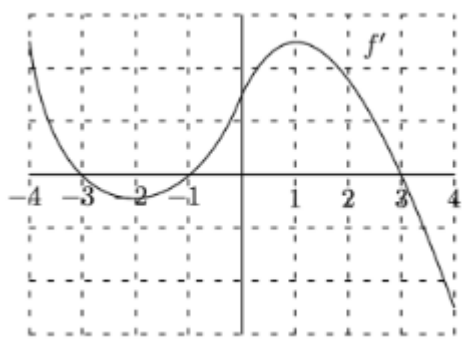
Which of the following are true about the function f if its derivative is $f'(x) = (x-1)^2(4-x)$?

I. f is decreasing for all $x < 4$.
II. f has a local maximum at $x = 1$.
III. f is concave up for all $1 < x < 3$.

A. I only
B. II only
C. III only
D. II and III only

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus



The graph of the derivative of a function f is shown above. Which of the following are true about the original function f ?

I. f is increasing on the interval $(-2, 1)$.
II. f is continuous at $x = 0$.
III. f has an inflection point at $x = -2$.

A. I only
B. II only
C. III only
D. II and III only

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Given that the function f is continuous at the point ($c, f(c)$), determine which of the following statements could be false.

A. $\lim_{x \to c} f(x)$ exists.
B. $\lim_{x \to c} f(x) = f(c)$.
C. $\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x)$.
D. $\lim_{x \to c} f(x) = c$.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

If $f(x) = 2 + |x - 3|$ for all x, then the value of the derivative $f'(x)$ at $x = 3$ is

A. -1
B. 0
C. 1
D. nonexistent

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let $f : [a, b] \to \mathbb{R}$ be a differentiable function, and define $g(x) = (f(x))^2$.
Apply the Mean Value Theorem (MVT) to the function g on the interval $[a, b]$.
Then which of the following is correct?

A. There exists $c \in (a, b)$ such that

$$\frac{f(b)^2 - f(a)^2}{b - a} = f'(c)^2$$

B. There exists $c \in (a, b)$ such that

$$\frac{f(b)^2 - f(a)^2}{b - a} = 2f(c)f'(c)$$

C. There exists $c \in (a, b)$ such that

$$\frac{f(b)^2 - f(a)^2}{b - a} = f(c)^2$$

D. There exists $c \in (a, b)$ such that

$$\frac{f(b)^2 - f(a)^2}{b - a} = f'(c)[f(a) + f(b)]$$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

$\lim_{n \to \infty} \frac{2^{n+1} + 3^{n+1}}{2^n + 3^n}$ equals

A. 3

B. 2

C. 1

D. 0

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let

$$f(x) = (x-1)(x-2)(x-3)(x-4)(x-5)$$

The number of distinct real roots of the equation $\frac{d}{dx}f(x) = 0$ is exactly

A. 2
B. 3
C. 4
D. 5

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The positive integers m and n are relatively prime, and chosen in such a way that

$$\lim_{x \to 2} \frac{5\sqrt{1+4x} - 3\sqrt{1+12x}}{x-2} = -\frac{m}{n}.$$

What is $n - 4m$?

A. -1
B. 2
C. -3
D. 4

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

$$f(x) = \begin{cases} 2x-2 & \text{for } x < 3 \\ 2x-4 & \text{for } x \geq 3 \end{cases}$$

Let f be the piecewise-linear function defined above. Which of the following statements are true?

I. $\lim_{h \to 0^-} \frac{f(3+h)-f(3)}{h} = 2$

II. $\lim_{h \to 0^+} \frac{f(3+h)-f(3)}{h} = 2$

III. $f'(3) = 2$

A. None

B. II only
C. I and II only
D. I, II, and III

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

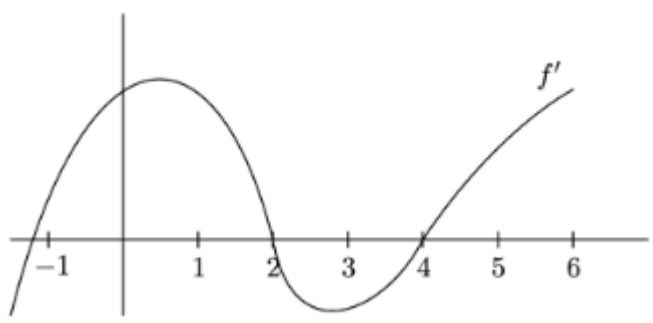
Which of the following statements is always true for a function $f(x)$?

i. If $f(x)$ and $g(x)$ are continuous at $x = a$, then $\frac{f(x)}{g(x)}$ is continuous at $x = a$.
ii. If $f(x) + g(x)$ is continuous at $x = a$ and $f'(a) = 0$, then $g(x)$ is continuous ta $x = a$.
iii. If $f(x) + g(x)$ is differentiable at $x = a$, then $f(x)$ and $g(x)$ are both differentiable at $x = a$

A. only i.
B. only ii.
C. only iii.
D. i. and ii.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus



The graph of the first derivative f' for a function f is shown above. Which of the following statements must be true?

I. $f(4) = f(-1)$
II. $f(2) > f(4)$
III. $f''(2) < f''(4)$

A. I only
B. II only
C. II and III only
D. I and III only

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If f is continuous at $x = 2$, and if f is defined as

$$f(x) = \begin{cases} \frac{\ln x^2 - x \ln x}{x-2} & x \neq 2 \\ k & x = 2, \end{cases}$$

then k is

A. -1
B. -2
C. $-\frac{e}{4}$
D. $-\ln 2$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If $f'(x)$ and $g'(x)$ exist and $f'(x) > g'(x)$ for all real x, then the graph of $y = f(x)$ and the graph of $y = g(x)$

A. intersect exactly once.
B. intersect no more than once.
C. do not intersect.
D. could intersect more than once.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

$\lim_{n\to\infty} \frac{1-2+3-4+5-6+\ldots+(-2n)}{\sqrt{n^2+1}+\sqrt{n^2-1}}$ equals

A. ∞
B. $1/2$
C. 0
D. $-1/2$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let $f(x, y) = \frac{ax^2+by^2}{xy}$, where a and b are constants. If $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}$ at $x = 1$ and $y = 2$, then the relation between a and b is:

A. $a = \frac{b}{4}$

B. $a = \frac{b}{2}$

C. $a = 2b$

D. $a = 4b$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let

$$f(x,y) = 15 - x^2 + 2y^2 + 6x - 8y.$$

Which of the following statements is correct about the critical point(s) of $f(x,y)$?

A. The function has a critical point at $(3,2)$, which is a local maximum.
B. The function has a critical point at $(3,2)$, which is a local minimum.
C. The function has no critical points.
D. The function has a critical point at $(3,2)$, which is a saddle point.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Consider the function $f(x) = \sqrt{3-x}$. Which of the following is the definition of $f'(-1)$?

A. $f'(-1) = \lim_{x\to -1} \frac{\sqrt{3-x}-2}{x+1}$

B. $f'(-1) = \lim_{x\to \infty} \frac{\sqrt{3-x}-2}{x+1}$

C. $f'(-1) = \lim_{h\to 0} \frac{\sqrt{3-x-h}-\sqrt{3-x}}{h}$

D. $f'(-1) = \lim_{h\to 0} \frac{h}{\sqrt{3-x-h}-\sqrt{3-x}}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

The function f is defined on the closed interval $[2,4]$ and $f(2) = f(3) = f(4)$. On the open interval $(2,4)$, f is continuous and strictly decreasing. Which of the following statements is true

A. f attains neither a minimum value nor a maximum value on the closed interval $[2,4]$.
B. f attains a minimum value but does not attain a maximum value on the closed interval $[2,4]$.
C. f attains a maximum value but does not attain a minimum value on the closed interval $[2,4]$.
D. f attains both a minimum value and a maximum value on the closed interval $[2,4]$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

x	0	1	2
$f(x)$	1	k	2

The function f is continuous on the closed interval $[0, 2]$ and has values that are given in the table above. The equation $f(x) = \frac{1}{2}$ must have at least two solutions in the interval $[0, 2]$ if $k =$

A. 0

B. $\frac{1}{2}$

C. 1

D. 2

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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